CCTV Exam Intelligence Event & Camera API

Introduction & Requirements

This API enables real-time event ingestion and camera metadata access for AI-based exam monitoring systems across the CCTV Exam Intelligence platform. Use it to retrieve live camera feeds, status updates, and track core AI system enumerations.

Important Security Requirement: All requests must be made over secure HTTP/HTTPS endpoints with a valid API credential key. Requests without authentic headers are explicitly rejected.

Authentication Setup

Mechanism: Authentication uses API Key verification generated uniquely per active exam schedule window.

Required Header: Authorization: Bearer <API_KEY>

Example Request via Curl:

```
curl -H "Authorization: Bearer YOUR_API_KEY"
http://localhost/api/v1/master/camera-typessuper
```

Global Route Map Matrix

Method	Resource URL Path	Service Operational Mapping Target
GET	/api/v1/cameras	Fetch all operational cameras with location and status metadata dictionaries
GET	/api/v1/master/camera-types	Reference data enumeration list of system camera type configurations
GET	/api/v1/master/camera-statuses	Reference validation data for system connectivity status states
GET	/api/v1/master/event-types	Reference catalog list mapping tracking IDs to AI event descriptions
GET	/api/v1/master/current-schedule	Active production exam window

		schedule metadata with assigned Client fields
POST	/api/v1/camera-live-status	Webhook Trigger: Live online/offline heartbeat status changes at a test centre
POST	/api/v1/events	Webhook Target: Real-time telemetry feed delivering streaming AI core violations

1. Live Telemetry Data Retrieval APIs

Get Cameras Dataset

Route: GET /api/v1/cameras

Fetch structural tracking details of all active camera arrays bound to configured testing centres.

Response Payload Schema Blueprint:


```
{
  "cameras": [
    {
      "cam_id": "1000736$1$0$5-d14.camview.io",
      "cam_type": 1,
      "cam_type_name": "InsideLabs",
      "tc_code": "100104",
      "center_name": "BALAJI DIGITAL ZONE",
      "city": "DELHI",
      "state": "Delhi",
      "status": 1,
      "status_name": "AnalyticsLive",
      "vendor_id": ""
    }
  ]
}
```


Field Key	Data Type	Schema Mapping Definition Description
cam_id	string	Primary Key identifier hash applied uniquely to the capture hardware line.
cam_type	integer	Structural look-up index key referencing core master types

		catalog dictionary.
cam_type_name	string	Human-legible clear name descriptive of the physically deployed placement location.
tc_code	string	Operational center unique tracking reference alphanumeric string code.
center_name	string	Descriptive physical title tag designated for the local test site facilities.
city / state	string	Geographic localization data capturing the city and operational regional state borders.
status	integer	Connectivity workflow status index mapping to active reference state variables.
status_name	string	Readable string tag mirroring the physical operational system status category state.
vendor_id	string	Tracking parameter metadata string tracking allocated hardware service vendor lines.

2. Master Structural Reference Configurations

Camera Type Enumeration Catalog

Endpoint Route: GET /api/v1/master/camera-types


```
{
  "camera_types": [
    { "id": 1, "name": "InsideLabs", "vendor_id": "" },
    { "id": 2, "name": "ServerRoom", "vendor_id": "" },
    { "id": 4, "name": "Outside", "vendor_id": "" },
    { "id": 5, "name": "EntryExit", "vendor_id": "" },
    { "id": 9, "name": "Invalid", "vendor_id": "" },
```

```
    { "id": 0, "name": "Other", "vendor_id": "" }
  ]
}
```

Camera Stream Connection Status Scaffolds

Endpoint Route: GET /api/v1/master/camera-statuses

```
{
  "statuses": [
    { "id": 0, "name": "Idle/Ready", "vendor_id": "" },
    { "id": 1, "name": "AnalyticsLive", "vendor_id": "" },
    { "id": 100, "name": "ClassificationPending", "vendor_id": "" },
    { "id": 200, "name": "Excluded", "vendor_id": "" },
    { "id": 9999, "name": "Other", "vendor_id": "" }
  ]
}
```

AI Event Incident Model Classification Index

Endpoint Route: GET /api/v1/master/event-types

```
{
  "event_types": [
    { "id": 12, "name": "MobileDetectioninServerRoom", "vendor_id": "" },
    { "id": 13, "name": "MobileDetectioninInsideLab", "vendor_id": "" },
    { "id": 15, "name": "ImproperFurniture", "vendor_id": "" },
    { "id": 34, "name": "MovementatEntryExit", "vendor_id": "" },
    { "id": 32, "name": "ServerRoomThreshold", "vendor_id": "" },
    { "id": 11, "name": "InvigilatorNotMoving", "vendor_id": "" },
    { "id": 35, "name": "IntruderDetection", "vendor_id": "" },
    { "id": 33, "name": "CloseProximity", "vendor_id": "" },
    { "id": 31, "name": "CrowdDetectionEntryExit", "vendor_id": "" },
    { "id": 30, "name": "CrowdDetectionOutside", "vendor_id": "" },
    { "id": 43, "name": "CameraFeedInverted", "vendor_id": "" },
    { "id": 42, "name": "0Candidatesinaroom", "vendor_id": "" },
    { "id": 41, "name": "Lessthan5Candidatesinaroom", "vendor_id": "" }
  ]
}
```

Current Exam Scheduling Matrix

Endpoint Route: GET /api/v1/master/current-schedule

```
{
  "schedules": [
    { "client_name": "SSC", "schedule_name": "CTGD-Apr2026", "vendor_id": "" }
  ]
}
```

3. Real-Time Distributed Webhook Streaming Frameworks

Live Camera Status Synchronization Channel

Trigger Request Method: POST /api/v1/camera-live-status

URL Queries: tc_code (String, Required Query Attribute Target Parameters)

Behavior Model: CCTV streams an automated outbound POST payload tracking comprehensive center-wide hardware connectivity array configurations whenever state changes happen.

Pushed Real-Time Multi-Array Event Body Frame JSON Structure:


```
[
  {
    "tc_code": "2288",
    "exam_code": "XX",
    "client_name": "SSC",
    "schedule_name": "CTGD-Apr2026",
    "vendor_id": "",
    "cameras": [
      { "cam_id": "202288$1$0$62502", "status": "online", "last_seen": "2026-06-
25T18:00:58+05:30" },
      { "cam_id": "202288$1$0$62415", "status": "offline", "last_seen": "2026-06-
25T17:55:12+05:30" }
    ]
  },
  {
    "tc_code": "2288",
    "exam_code": "XX",
    "client_name": "SSC",
    "schedule_name": "CTGD-Apr2026",
    "exam_id": 123,
    "cameras": [
      { "cam_id": "202288$1$0$62502", "status": "online", "last_seen": "2026-07-
08T18:00:58+05:30" },
      { "cam_id": "202288$1$0$62415", "status": "offline", "last_seen": "2026-07-
08T17:55:12+05:30" }
    ]
  }
]
```


Post Events Violation Stream Channel

Trigger Request Method: POST /api/v1/events

Description: Pushes high-priority operational computer vision telemetry alert incidents as structured object arrays down to the client gateway instantly.

Streaming Payload Event Ingestion Envelope Model:


```
{
  "events": [
    {
```

```
      "event_id": 2495393,
      "cam_id": "1000113$1$0$19-d15.camview.io",
      "cam_type": 1,
      "event_type": 11,
      "event_name": "Invigilator not moving",
      "timestamp": "2026-09-24T10:59:29",
      "image_base64": "/9j/4AAQSkZJRgABAQAAAQABAAD...//2Q==",
      "image_url": "",
      "image_name": "",
      "schedule_name": "SPE-Sep2026",
      "shift": "shift2",
      "vendor_id": ""
    }
  ]
}
```


4. Gateways Response Status Codes Dictionary

Exception Standard: All core systems and error frames fallback into a singular envelope structure for uniform exception handling logic structures:


```
{ "error": "Human-readable explanation message reporting system level exception faults." }
```


Status	Gateway Meaning	Infrastructure Condition Handling Context Mapping
200	OK	Action completely validated, processed, and confirmed successfully.
400	Bad Request	Structural formatting errors, missing required properties, or invalid elements.
401	Unauthorized	Security identity missing, credentials malformed, or invalid tokens.
402	Request Failed	Correct schema parsing context found but transaction validation rules logic failed.
403	Forbidden	Validated credential check verified successfully but permissions profile holds non-authorized

		scopes.
404	Not Found	Target system URL path context lookup failed to resolve an active asset resource matching rules.
405	Method Not Allowed	Route location matches but the HTTP methodology verb is non-supported by the active configuration.
409	Conflict	Request parameters directly intersect with locked records inside current relational DB structures.
424	Dependency Failed	Upstream computer vision microservice node dropped connections or responded with structural errors.
429	Too Many Requests	Active throttling execution limit triggered. Incoming streams must gracefully execute retry delays.
500+	Server Error	Infrastructure runtime exception crashed internal background processes ungracefully.
502	Bad Gateway	Edge proxies intercept invalid response signatures from down-level background application nodes.